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ECEN 300: PS12

7.27(a), 7.31(a), 7.32(b), 7.35(a)(b), 7.42(c), MATLAB 7.58

7.27 (a) Determine the impulse response corresponding to the following transfer function if:

(i) the system is stable &

(ii) the system is causal

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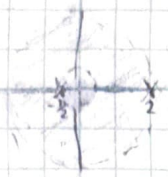
(a) $H(z) = \dfrac{2 - \frac{3}{2}z^{-1}}{(1-2z^{-1})(1+\frac{1}{2}z^{-1})}$

poles: $-\frac{1}{2}, 2$

(i) stable: include unit circle



first: $|z| < 2$

second: $|z| > \frac{1}{2}$

together: $\frac{1}{2} < |z| < 2$

partial frac: $\dfrac{A}{1-2z^{-1}} + \dfrac{B}{1+\frac{1}{2}z^{-1}} = \dfrac{2-\frac{3}{2}z^{-1}}{(1-2z^{-1})(1+\frac{1}{2}z^{-1})} = \underbrace{\dfrac{1}{1-2z^{-1}}}_{|z|<2} + \underbrace{\dfrac{1}{1+\frac{1}{2}z^{-1}}}_{|z|>\frac{1}{2}}$

$A(1+\frac{1}{2}z^{-1}) + B(1-2z^{-1}) = 2 - \frac{3}{2}z^{-1}$

$A(1+\frac{1}{2}x) + B(1-2x) = 2 - \frac{3}{2}x$

$A + \frac{1}{2}Ax + B - 2Bx = 2 - \frac{3}{2}x$

$A + B = 2$

$\frac{1}{2}A - 2B = -\frac{3}{2}$

$\begin{bmatrix} 1 & 1 & 2 \\ \frac{1}{2} & -2 & -\frac{3}{2} \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & \cdot \\ 0 & 1 & \cdot \end{bmatrix}$

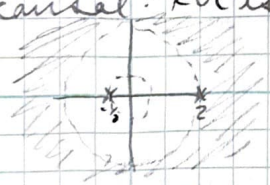
E.1.1: $\mathcal{Z}^{-1}\{-\alpha^n u[-n-1]\} = \dfrac{1}{1-\alpha z^{-1}}$ ROC: $|z| < |\alpha|$

E.1: $\mathcal{Z}^{-1}\{\alpha^n u[n]\} = \dfrac{1}{1-\alpha z^{-1}}$, ROC $= |z| > \alpha$

$$\boxed{h[n] = -2^n u[-n-1] + (-\tfrac{1}{2})^n u[n]}$$

* notecard: transfer is z-transform $H(z)$ of the impulse response $h[n]$

(ii) causal: ROC is expanding



first: $|z| > 2$

second: $|z| > \frac{1}{2}$

together: $|z| > 2$

$$\boxed{h[n] = (2)^n u[n] + (-\tfrac{1}{2})^n u[n]}$$

7.31 (a) Determine (i) transfer function and
(ii) impulse response representations of the causal systems described by the following difference equations.

(a) $y[n] - \frac{1}{2}y[n-1] = 2x[n-1]$

E.2 $\mathcal{Z}\{y[n-k]\} = z^{-k}X(z)$

(i) $Y(z) - \frac{1}{2}z^{-1}Y(z) = 2z^{-1}X(z)$

$Y(z)[1 - \frac{1}{2}z^{-1}] = 2z^{-1}X(z)$

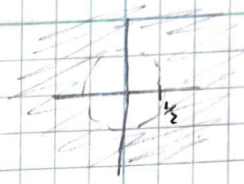
$H(z) = \frac{Y(z)}{X(z)} = \frac{2z^{-1}}{1-\frac{1}{2}z^{-1}}$

$$\boxed{H(z) = \frac{2z^{-1}}{1-\frac{1}{2}z^{-1}}}$$ ← transfer function

(ii) causal : ROC is expanding

poles : $|z| = \frac{1}{2}$

ROC $= |z| > \frac{1}{2}$



$H(z) = 2z^{-1} \cdot \underbrace{\frac{1}{1-\frac{1}{2}z^{-1}}}_{= A(z)}$

$\mathcal{Z}^{-1}\{\ \} = (\frac{1}{2})^n u[n] = a[n]$

$\mathcal{Z}^{-1}\{2z^{-1}A(z)\} = 2a[n-1] = 2[(\frac{1}{2})^{n-1}u[n-1]]$

$$\boxed{h[n] = 2(\tfrac{1}{2})^{n-1}u[n-1]}$$ ← impulse response

7.32(b) Determine (i) transfer function & (ii) difference equation from impulse response:

(b) $h[n] = (\frac{1}{3})^n u[n] + (\frac{1}{2})^{n-2} u[n-1]$

E.1 : $\mathcal{Z}\{\alpha^n u[n]\} = \frac{1}{1-\alpha z^{-1}}$, $|z|>|\alpha|$

E.2 : $\mathcal{Z}\{x[n-k]\} = z^{-k}X(z)$, R_x except possibly $|z| = 0, \infty$

$$h[n] = (\tfrac{1}{3})^n u[n] + (\tfrac{1}{2})^{-1} \cdot (\tfrac{1}{2})^{n-1} u[n-1]$$

$$= \underbrace{(\tfrac{1}{3})^n u[n]}_{a[n]} + 2\underbrace{(\tfrac{1}{2})^{n-1} u[n-1]}_{b[n]} \qquad \underbrace{(\tfrac{1}{2})^n u[n]}_{c[n]}$$

$$A(z) = \frac{1}{1-\frac{1}{3}z^{-1}} \; ; \; C(z) = \frac{1}{1-\frac{1}{2}z^{-1}}$$

$$b[n] = 2c[n-1]$$

$$B(z) = 2z^{-1} \cdot \frac{1}{1-\frac{1}{2}z^{-1}}$$

$$\boxed{H(z) = \frac{1}{1-\frac{1}{3}z^{-1}} + \frac{2z^{-1}}{1-\frac{1}{2}z^{-1}}}$$ transfer function

(ii) $$H(z) = \frac{(1-\frac{1}{2}z^{-1})}{(1-\frac{1}{3}z^{-1})(1-\frac{1}{2}z^{-1})} + \frac{2z^{-1}(1-\frac{1}{3}z^{-1})}{(1-\frac{1}{3}z^{-1})(1-\frac{1}{2}z^{-1})}$$

$$= \frac{1-\frac{1}{2}z^{-1} + 2z^{-1} - \frac{2}{3}z^{-2}}{1-\frac{1}{2}z^{-1}-\frac{1}{3}z^{-1}+\frac{1}{6}z^{-2}} = \frac{1+\frac{3}{2}z^{-1}-\frac{2}{3}z^{-2}}{1+\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}} = \frac{Y(z)}{X(z)}$$

$$Y(z)\left[1+\tfrac{5}{6}z^{-1}+\tfrac{1}{6}z^{-2}\right] = X(z)\left[1+\tfrac{3}{2}z^{-1}-\tfrac{2}{3}z^{-2}\right]$$

$$\boxed{y[n] - \tfrac{5}{6}y[n-1] + \tfrac{1}{6}y[n-2] = x[n] + \tfrac{3}{2}x[n-1] - \tfrac{2}{3}x[n-2]}$$

difference equation

7.35 (a) For each of the following systems, identify the transfer function of the inverse system, and determine whether the inverse system can be both causal and stable:

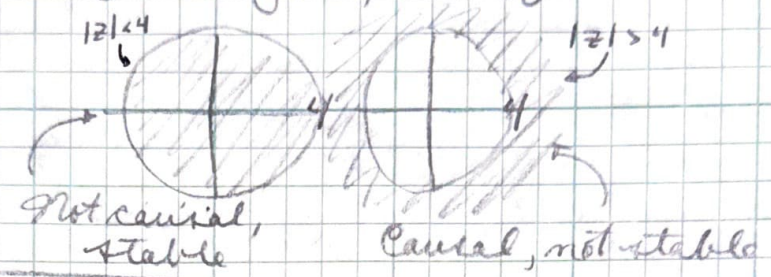
$$(a)\quad H(z) = \frac{1 - 8z^{-1} + 16z^{-2}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$$

$$H^{inv}(z) = \frac{1}{H(z)} = \boxed{\frac{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}{1 - 8z^{-1} + 16z^{-2}}}$$

$* z^{-1} = x^{(k-n-p)}$

$$\frac{\frac{1}{4}x^2 - \frac{1}{2}x + 1}{16x^2 - 8x + 1} = \frac{\frac{1}{4}x^2 - \frac{1}{2}x + 1}{16(x - \frac{1}{4})^2} = \frac{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}{16(z^{-1} - \frac{1}{4})^2}$$

poles: $z = 4$ with this pole, two options:



This system can't be both causal & stable.

7.35 (b) Same as above, with:

$$H(z) = \frac{z^2 - \frac{81}{100}}{z^2 - 1}$$

$$\boxed{H^{inv}(z) = \frac{z^2 - 1}{z^2 - \frac{81}{100}}} = \frac{z^2 - 1}{(z - \frac{9}{10})^2} \quad \text{poles: } z = \frac{9}{10}$$

Two options:

It is possible for this system to be both causal and stable, if the ROC is $|z| > \frac{9}{10}$. This ROC includes the unit circle and is expanding.

7.42 (c) Use the unilateral z-transform to determine the zero-state response, the natural response, and the complete response of the system described by the following difference equation with the given input & initial conditions:

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$$(c)\ y[n] - \tfrac{1}{4}y[n-1] - \tfrac{1}{8}y[n-2] = x[n] + x[n-1];$$

$$y[-1] = 1;\ y[-2] = -1;\ x[n] = 3^n u[n]$$

E.2.2:

$$x[n-k] \xleftrightarrow{\mathcal{Z}u} X[-k] + x[-k+1]z^{-1} + \dots + x[-1]z^{-k+1} + z^{-k}X(z) \quad \text{For } k>0$$

$$Y[n] - \tfrac{1}{4}\left[y[-1] + z^{-1}Y(z)\right] - \tfrac{1}{8}\left[y[-2] + y[-1]z^{-1} + z^{-2}Y(z)\right]$$

$$= \frac{1}{1-3z^{-1}} + \frac{z^{-1}}{1-3z^{-1}}$$

$$Y(z) - \tfrac{1}{4}\overbrace{y[-1]}^{1} + \tfrac{1}{4}Y(z)z^{-1} - \tfrac{1}{8}\overbrace{y[-2]}^{-1} + \tfrac{1}{8}y[-1]z^{-1} + \tfrac{1}{8}Y(z)z^{-2} = \frac{1+z^{-1}}{1-3z^{-1}}$$

$$Y(z) - \tfrac{1}{4} + \tfrac{1}{4}Y(z)z^{-1} + \tfrac{1}{8} + \tfrac{1}{8}z^{-1} + \tfrac{1}{8}Y(z)z^{-2} = \frac{1+z^{-1}}{1-3z^{-1}}$$

$$Y(z) - \tfrac{1}{4}Y(z)z^{-1} + \tfrac{1}{8}Y(z)z^{-1} = \tfrac{1}{4} - \tfrac{1}{8} - \tfrac{1}{8}z^{-1} + \frac{1+z^{-1}}{1-3z^{-1}}$$

$$Y(z)\left[1 - \tfrac{1}{4}z^{-1} + \tfrac{1}{8}z^{-2}\right] = \tfrac{1}{4} - \tfrac{1}{8} - \tfrac{1}{8}z^{-1} + \frac{1+z^{-1}}{1-3z^{-1}}$$

$$Y(z) = \frac{\tfrac{1}{4} - \tfrac{1}{8} - \tfrac{1}{8}z^{-1} + \frac{1+z^{-1}}{1-3z^{-1}}}{1 - \tfrac{1}{4}z^{-1} + \tfrac{1}{8}z^{-2}} = \frac{-\tfrac{1}{8} - \tfrac{1}{8}z^{-1} + \frac{1+z^{-1}}{1-3z^{-1}}}{1 + \tfrac{1}{4}z^{-1} + \tfrac{1}{8}z^{-2}}$$

$$= \frac{-\tfrac{1}{8} - \tfrac{1}{8}z^{-1} + \frac{1+z^{-1}}{1-3z^{-1}}}{(1-\tfrac{1}{2}z^{-1})(1+\tfrac{1}{4}z^{-1})} = \frac{-\tfrac{1}{8}}{(1-\tfrac{1}{2}z^{-1})(1+\tfrac{1}{4}z^{-1})} + \frac{-\tfrac{1}{8}z^{-1}}{(1-\tfrac{1}{2}z^{-1})(1+\tfrac{1}{4}z^{-1})} + \frac{1+z^{-1}}{(1-3z^{-1})}$$

$$\frac{A}{1-\tfrac{1}{2}z^{-1}} + \frac{B}{1+\tfrac{1}{4}z^{-1}} = \frac{-\tfrac{1}{8} - \tfrac{1}{8}z^{-1}}{(1-\tfrac{1}{2}z^{-1})(1+\tfrac{1}{4}z^{-1})}$$

$$A(1+\tfrac{1}{4}z^{-1}) + B(1-\tfrac{1}{2}z^{-1}) = -\tfrac{1}{8} - \tfrac{1}{8}z^{-1}$$

$$A + \tfrac{1}{4}Az^{-1} + B - \tfrac{1}{2}Bz^{-1} = -\tfrac{1}{8} - \tfrac{1}{8}z^{-1}$$

$$A + B = -\tfrac{1}{8}$$

$$\tfrac{1}{4}A - \tfrac{1}{2}B = -\tfrac{1}{8}$$

$$\begin{bmatrix} 1 & 1 & -\tfrac{1}{8} \\ \tfrac{1}{4} & -\tfrac{1}{2} & -\tfrac{1}{8} \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & -\tfrac{1}{4} \\ 0 & 1 & \tfrac{1}{8} \end{bmatrix}$$

$$\frac{A}{1-\tfrac{1}{2}z^{-1}} + \frac{B}{1+\tfrac{1}{4}z^{-1}} + \frac{C}{1-3z^{-1}} = \frac{1+z^{-1}}{(\ )(\ )}$$

$$A(1+\tfrac{1}{4}z^{-1})(1-3z^{-1}) + B(1-\tfrac{1}{2}z^{-1})(1-3z^{-1}) + C(1-\tfrac{1}{2}z^{-1})(1+\tfrac{1}{4}z^{-1}) = 1+z^{-1}$$

$$A(1 + -\tfrac{1}{4}z^{-1} - \tfrac{3}{4}z^{-2})$$

$$B(1 - \tfrac{7}{2}z^{-1} + \tfrac{3}{2}z^{-2})$$

$$C(1 - \tfrac{1}{4}z^{-1} - \tfrac{1}{8}z^{-2}) = 1 + z^{-1}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ -\tfrac{1}{4} & -\tfrac{7}{2} & -\tfrac{1}{4} & 1 \\ -\tfrac{3}{4} & \tfrac{3}{2} & -\tfrac{1}{8} & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a & -2/5 \\ a & -1/13 \\ a & 96/65 \end{bmatrix}$$

$$Y(z) = \frac{-1/4}{1-\frac{1}{2}z^{-1}} + \frac{1/8}{1+\frac{1}{4}z^{-1}} + \frac{96/65}{1-3z^{-1}} - \frac{2/5}{1-\frac{1}{2}z^{-1}} - \frac{1/13}{1+\frac{1}{4}z^{-1}}$$

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$$\boxed{y[n] = \underbrace{-\tfrac{1}{4}\left(\tfrac{1}{2}\right)^n u[n] + \tfrac{1}{8}\left(-\tfrac{1}{4}\right)^n u[n]}_{\text{forced response}} + \underbrace{\tfrac{96}{65}(3)^n u[n] - \tfrac{2}{5}\left(\tfrac{1}{2}\right)^n u[n] - \tfrac{1}{13}\left(-\tfrac{1}{4}\right)^n u[n]}_{\text{natural response}}}$$

complete response

MATLAB Problem 7.58

Use the MATLAB command **zplane** to solve Problems 7.35(a) and (b).

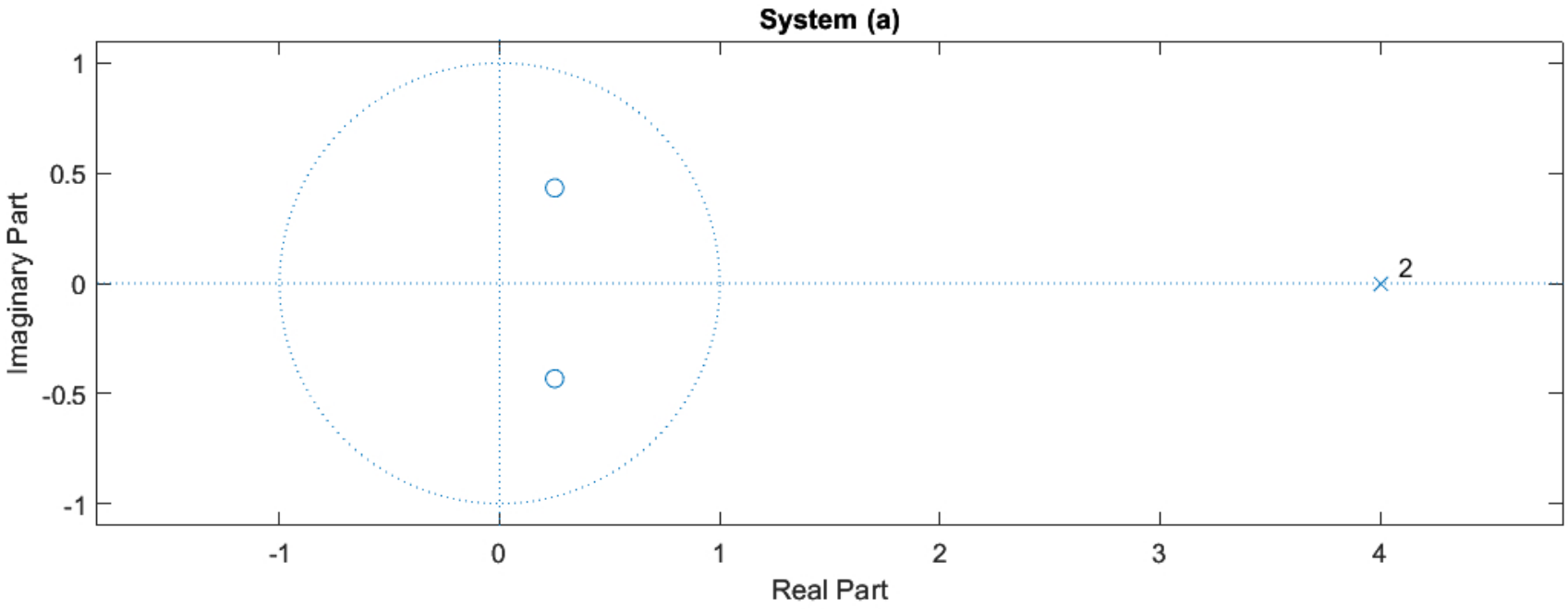
For each of the following systems, identify the transfer function of the inverse system, and determine whether the inverse system can be both causal and stable:

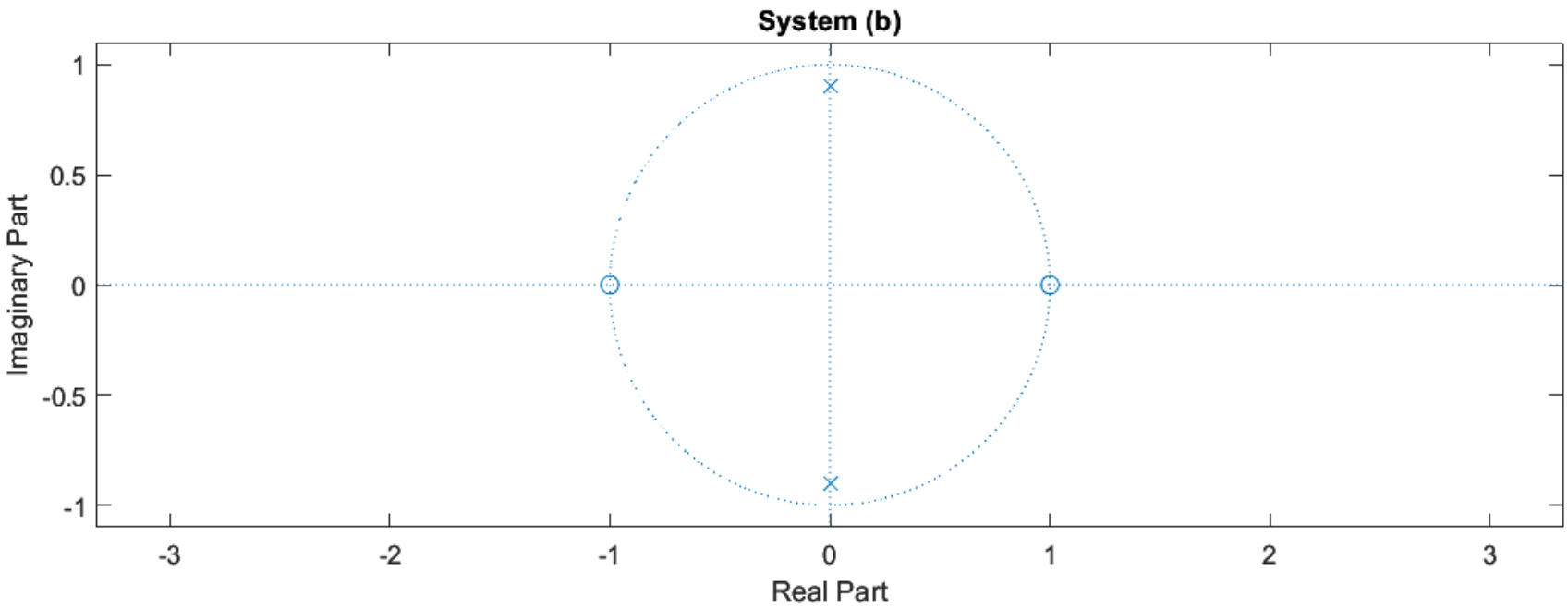
(a) $H(z) = \dfrac{1 - 8z^{-1} + 16z^{-2}}{1 - 1/2z^{-1} + 1/4z^{-2}}$

(b) $H(z) = \dfrac{z^2 - 81/100}{z^2 - 1}$

```
subplot(2,1,1)
b = [1 -1/2 1/4];
a = [1 -8 16];
zplane(b, a)
title('System (a)')

subplot(2,1,2)
b = [1 0 -1];
a = [1 0 81/100];
zplane(b, a)
title('System (b)')
```


System (a)
Imaginary Part
Real Part
1
0.5
0
-0.5
-1
-1
0
1
2
3
4
2


System (b)
Imaginary Part
Real Part
1
0.5
0
-0.5
-1
-3
-2
-1
0
1
2
3